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Partitions and their generating functions

DEFN An (integer) partition $\lambda = (\lambda_1, \lambda_2, \lambda_3, \dots)$ of n is a • weakly decreasing, eventually zero, sequence of nonnegative integers with $\lambda_1 + \lambda_2 + \lambda_3 + \dots = n$.

We write $\lambda \vdash n$ (' λ divides n ') and $|\lambda| = n$. ← 'size' of λ

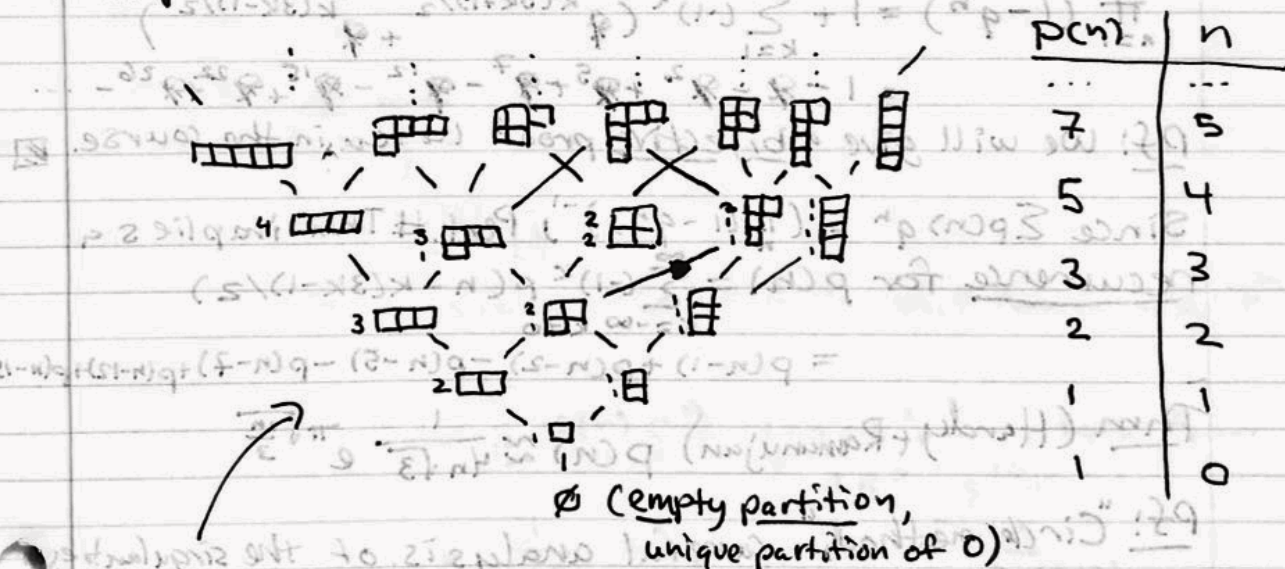
e.g. $\lambda = (5, 5, 3, 1, 0, 0, \dots) = (5, 5, 3, 1, 0) = (5, 5, 3, 1) \vdash 14 = 5+5+3+1$

Its length $l(\lambda) := \#\{i: \lambda_i > 0\} = \# \text{ of nonzero parts } \lambda_i$.

Its Young diagram is a left + top justified array of boxes, with λ_i boxes in the i th row from the top.

e.g. $\lambda = (5, 5, 3, 1) \leftrightarrow$ 

Let $p(n) := \# \text{ of partitions } \lambda \vdash n$



$\mathbb{Y} =$ Young's lattice, the poset of all partitions, ordered by containment of Young diagrams
 $\mu \rightarrow \lambda$ means λ obtained from μ by adding one box

$$\sum_{n \geq 0} p(n) q^n = \sum_{\text{all partitions } \lambda} q^{|\lambda|} = \left[\begin{array}{cc} 1 + q + q^2 + q^3 + \dots & 1 + q^2 + q^4 + q^6 + \dots \\ 1 + x_1 + x_1^2 + x_1^3 + \dots & 1 + x_2 + x_2^2 + x_2^3 + \dots \\ (1 + \square + \square + \square + \dots) & (1 + \square + \square + \square + \dots) \\ 1 + x_1^3 + x_1^6 + \dots & 1 + x_4 + x_4^2 + \dots \\ (1 + \square + \square + \dots) & (1 + \square + \square + \dots) \end{array} \right]$$

$$+ \dots + \left[\begin{array}{c} x_1^4 x_2^3 x_3^2 x_4^1 \\ = x_1^2 x_2^3 x_3^0 x_4^1 = q^2 q^6 q^0 q^4 = q^{12} \end{array} \right]$$

$$\begin{aligned} x_i &\mapsto q^i = (1 + q + q^2 + \dots)(1 + q^2 + q^4 + \dots)(1 + q^3 + q^6 + \dots) \\ &= \frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \frac{1}{1-q^3} \cdot \dots \quad \text{a convergent product} \\ &= \prod_{n=1}^{\infty} \frac{1}{1-q^n} \leftarrow \text{g.f. for all partitions, as a product formula!} \end{aligned}$$

Cultural asides:

Thm (Euler's pentagonal number theorem) "pentagonal numbers"

$$\prod_{n=1}^{\infty} (1 - q^n) = 1 + \sum_{k=1}^{\infty} (-1)^k (q^{k(3k+1)/2} + q^{k(3k-1)/2})$$

$$= 1 - q - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

Pf: We will give a bijective proof later in the course. $\square$

Since $\sum p(n) q^n = (\prod (1 - q^n))^{-1}$ Pent. # Thm implies a recurrence for $p(n) = \sum_{k=-\infty, k \neq 0}^{\infty} (-1)^k p(n - k(3k-1)/2)$

$$= p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) - \dots$$

Thm (Hardy + Ramanujan) $p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi \sqrt{\frac{2n}{3}}}$

Pf: "Circle method," careful analysis of the singularities of $\prod \frac{1}{1-q^n}$ thought of as an analytic function. Like what we saw with rational g.f. of Fibonacci #s, but much more hardcore complex analysis. $\square$

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G.f.'s for restricted classes of partitions: at wot

Let $d(n) := \#$ of partitions of n into distinct parts.

n	$d(n)$	
0	1	$\emptyset$
1	1	$\square$
2	1	$\square \square$
3	2	$\square \square \square$ $\square \square$
4	2	$\square \square \square \square$ $\square \square \square$
5	3	$\square \square \square \square \square$ $\square \square \square \square$ $\square \square \square \square$

$$D(q) := \sum_{n \geq 0} d(n) q^n = (1+q)(1+q^2)(1+q^3)(1+q^4) \cdots = \prod_{j=1}^{\infty} (1+q^j)$$

Let $o(n) := \#$ partitions of n into odd parts.

n	$o(n)$	
0	1	$\emptyset$
1	1	$\square$
2	1	$\square \square$
3	2	$\square \square \square$ $\square \square$
4	2	$\square \square \square \square$ $\square \square \square$
5	3	$\square \square \square \square \square$ $\square \square \square \square$ $\square \square \square \square$

Looks like possibly $o(n) = d(n)$, but how to show it?

G.f.'s!

$$\begin{aligned} O(q) &:= \sum_{n \geq 0} o(n) q^n = (1+q+q^2+\cdots)(1+q^3+q^6+\cdots) \\ &\quad (1+q^5+q^{10}+\cdots) \cdots \\ &= \frac{1}{1-q} \cdot \frac{1}{1-q^3} \cdot \frac{1}{1-q^5} \cdots = \prod_{j=1}^{\infty} \frac{1}{1-q^{2j+1}} \\ &= \prod_{j=1}^{\infty} (1+q^j) \end{aligned}$$

How to show $D(q) = O(q)$?

Well, $D(q) = (1+q)(1+q^2)(1+q^3)\dots$

$$= \frac{(1-q^2)}{(1-q)} \frac{(1-q^2)^2}{(1-q^2)} \frac{(1-q^3)^2}{(1-q^3)} \dots$$

Recall: $(1+x)(1-x) = (1-x^2)$

$$= \frac{(1-q^2)(1-q^4)(1-q^6)(1-q^8)\dots}{(1-q)(1-q^2)(1-q^3)(1-q^4)\dots}$$

$$= \frac{1}{(1-q)(1-q^3)(1-q^5)\dots} = O(q)!$$

Was that manipulation ok? Yes! Thinking slightly differently...

Let $R(q) := (1-q)(1-q^3)(1-q^5)\dots = \frac{1}{O(q)} \in \mathbb{C}[[q]]$

Want to show $1 \stackrel{?}{=} D(q)R(q)$ in $\mathbb{C}[[q]]$

$$1 + 0 \cdot q + 0 \cdot q^2 + \dots$$

$$= (1+q)(1+q^2)(1+q^3)\dots (1-q)(1-q^3)(1-q^5)\dots$$

$$= \frac{(1+q)(1-q)}{(1-q^2)} \frac{(1+q^2)(1-q^2)}{(1-q^4)} \dots$$

$$= (1-q^4) \frac{(1+q^3)(1+q^4)\dots}{(1-q^2)} \frac{(1-q^3)(1-q^5)\dots}{(1-q^4)}$$

$$= (1-q^4)(1-q^6) \frac{(1+q^4)(1+q^5)\dots}{(1-q^2)} \frac{(1-q^5)(1-q^7)\dots}{(1-q^4)}$$

$$= 1 + 0 \cdot q + 0 \cdot q^2 + 0 \cdot q^3 + \dots$$

= ... et cetera

Note: $\exists$ bijective proof that $d(n) = o(n)$ as well

(See Stanley Prop. 1.8.5) Basic idea is binary expansion:

e.g. $\lambda = (9^5, 5^{12}, 3^2, 1^3) = (9^{2^0+2^2}, 5^{2^2+2^3}, 3^{2^1}, 1^{2^0+2^1})$

$O(n)$ $(9, 9, 9, 9, 9, 5, 5, \dots, 5, 3, 3, 1, 1, 1)$

$n=14$

$\leftrightarrow \mu = (9 \cdot 2^0, 9 \cdot 2^1, 5 \cdot 2^2, 5 \cdot 2^3, 3 \cdot 2^1, 1 \cdot 2^0, 1 \cdot 2^1)$

$= (9, 36, 20, 40, 6, 1, 2)$

(reorder)

$= (40, 36, 20, 9, 6, 2, 1) \in d(n)$

//

hinted at by this manipulation

Compositions and their generating functions

DEF'N A composition $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$ of n , denoted $\alpha \models n$, is a sequence of positive integers $\alpha_i \in \{1, 2, 3, \dots\}$ w/ $\alpha_1 + \alpha_2 + \dots + \alpha_k = n$. (Unlike a partition, the α_i need not be weakly decreasing.) As w/ partitions, we call α_i the parts of the composition.

e.g. $\alpha = (1, 4, 2, 4)$ is a composition of 11 into 4 parts.

Let $\bar{c}_k(n)$ denote # compositions of n into k parts and $\bar{c}(n)$ denote # compositions of n (into any # of parts).

Prop. $\sum_{n=0}^{\infty} \bar{c}_k(n) \cdot x^n = \left(\frac{1}{1-x} - 1\right)^k = \left(\frac{x}{1-x}\right)^k$

Pf: Note $\left(\frac{1}{1-x} - 1\right) = 1 + x + x^2 + x^3 + \dots - 1 = x + x^2 + x^3 + \dots$

Now use "picture-writing":

e.g. $k=3 \rightarrow (x + x^2 + x^3 + \dots)(x + x^2 + x^3 + \dots)(x + x^2 + x^3 + \dots)$

Cor. $\sum_{n=0}^{\infty} \bar{c}(n) \cdot x^n = 1 + \frac{x}{1-2x}$

Pf: Note that $\bar{c}(n) = \sum_{k=0}^{\infty} \bar{c}_k(n)$, so

$$\begin{aligned} \sum_{n=0}^{\infty} \bar{c}(n) \cdot x^n &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^{\infty} \bar{c}_k(n) \right) \cdot x^n = \sum_{k=0}^{\infty} \left(\sum_{n=0}^{\infty} \bar{c}_k(n) \cdot x^n \right) \\ &= \sum_{k=0}^{\infty} \left(\frac{x}{1-x} \right)^k = \frac{1}{1 - \frac{x}{1-x}} \\ &= \frac{1-x}{1-2x} = 1 + \frac{x}{1-2x} \end{aligned}$$

Cor. For $n \geq 1$, $\bar{c}(n) = 2^{n-1}$.

Pf. $1 + \frac{x}{1-2x} = 1 + \sum_{n \geq 0} 2^n \cdot x^{n+1} = 1 + \sum_{n \geq 1} 2^{n-1} x^n$ $\square$
 Then extract coeff. of x^n in previous cor.

Since this is such a simple formula, we could ask for a direct proof, not using generating functions. In fact...

Prop. The # of compositions of n into k parts is $\bar{c}_k(n) = \binom{n-1}{k-1}$.

Pf. Let's say a sequence $\alpha = (\alpha_1, \dots, \alpha_k)$ of nonnegative integers $\alpha_i \in \{0, 1, \dots\}$ w/ $\sum \alpha_i = n$ is a weak composition of n .

Claim # of weak compositions of n into k parts $= \binom{k+n-1}{n} = \binom{k+n-1}{n}$ $\leftarrow$ recall 'multichoose' #

Pf. Write a weak composition of n , e.g.
 $\alpha = (2, 0, 1, 3)$ using 'stars and bars'
 as $\alpha = **|1*|***$.

We saw before that these patterns are counted by $\binom{k+n-1}{n}$.

Finally, $\exists$ a bijection $\{\text{weak comp. of } n \text{ into } k \text{ parts}\} \leftrightarrow \{\text{(usual) comp. of } n+k \text{ into } k \text{ parts}\}$
 $(\alpha_1, \alpha_2, \dots, \alpha_k) \mapsto (\alpha_1+1, \alpha_2+1, \dots, \alpha_k+1)$.

Hence # comp. of n into k parts $= \binom{k+n-1}{n-k} = \binom{k+n-k-1}{n-k} = \binom{n-1}{n-k} = \binom{n-1}{k-1}$ $\square$

Cor. # comp. of n is $\bar{c}(n) = 2^{n-1}$ for any $n \geq 1$.

Pf. $\bar{c}(n) = \sum_{k=1}^n \binom{n-1}{k-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} = 2^{n-1}$ $\checkmark$ $\square$

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Some useful formal power series

Let's define and study some specific elements of $\mathbb{C}[[x]]$:

DEFN $e^x := \sum_{n \geq 0} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$\log(1+x) := \sum_{n \geq 1} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

$\forall \lambda \in \mathbb{C}, (1+x)^\lambda := \sum_{k \geq 0} \binom{\lambda}{k} x^k$
 where $\binom{\lambda}{k} := \frac{\lambda(\lambda-1)(\lambda-2)\dots(\lambda-(k-1))}{k!} \in \mathbb{C}$ "generalized binomial coefficient"

(just like for $n \in \mathbb{N}$, $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots(n-(k-1))}{k!}$)

These formal power series satisfy the properties you'd expect:

E.g. ① $(1+x)^\lambda (1+x)^\mu = (1+x)^{\lambda+\mu} \in \mathbb{C}[[x]] \quad \forall \lambda, \mu \in \mathbb{C}$

② $e^x e^y = e^{x+y} \in \mathbb{C}[[x, y]]$

③ $e^{\log(1+x)} = 1+x, \quad e + \dots$

defined to be $= 1 + \log(1+x) + \frac{(\log(1+x))^2}{2!} + \dots$
 $= 1 + (x - \frac{x^2}{2} + \frac{x^3}{3} - \dots) + \frac{(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots)^2}{2} + \dots$

Why does this even converge in $\mathbb{C}[[x]]$?

Prop. If $A(x) = \sum_{n \geq 0} a_n x^n$, $B(x) = \sum_{n \geq 0} b_n x^n$, and $b_0 = 0$, then $A(B(x)) := \sum_{n \geq 0} a_n (B(x))^n$ converges in $\mathbb{C}[[x]]$.

How to justify ①, ②, ③, etc...? Could do a tedious manipulation of coefficients, but instead, since $e^x, \log(1+x), (1+x)^\lambda$ are also analytic functions we are familiar with (whose power series expansions are as above), we can use a trick from complex analysis...

standard fact from complex analysis, true under weaker hypotheses too...

Thm If $f(z) = \sum_{n \geq 0} a_n z^n$ is analytic for $|z| < R$ for some $R > 0$, and f vanishes on $|z| < R$, then $a_0 = a_1 = a_2 = \dots = 0$.

④ for $n \in \mathbb{N}$, $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \sum_{k=0}^n \binom{n}{k} x^k$

but also $\frac{1}{(1-x)^n} = (1+(-x))^{-n} = \sum_{k=0}^{\infty} \frac{(-n)(-n-1)(-n-2)\dots(-n-(k-1))}{k!} (-x)^k$

$(1+x+x^2+\dots)(1+x+x^2+\dots)\dots(1+x+x^2+\dots)$ (n terms in product, as if n=18 flavors of bagels)

$\sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k$

PS that $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$. ('n multichoose k') $\binom{n+k-1}{k} := \#$ k element multisets w/ entries in $\{1, 2, \dots, n\}$

'Stars and bars'

Stars indicate how many of each element is chosen, bars separate bins for each element

e.g. $\{1, 1, 2, 4, 4, 7\}$ is a 7-element multisubset of $\{1, 2, \dots, 9\}$

⑤ $\frac{1}{1-4x} = \sum_{k=0}^{\infty} \binom{-1}{k} (-4x)^k = \sum_{k=0}^{\infty} \binom{1+k-1}{k} 4^k x^k = \sum_{k=0}^{\infty} 4^k x^k \checkmark$

but also $\frac{1}{(1-4x)^2} = \sum_{k=0}^{\infty} \binom{2+k-1}{k} 4^k x^k = \sum_{k=0}^{\infty} (k+1) 4^k x^k$

$\frac{1}{(1-4x)^3} = \sum_{k=0}^{\infty} 4^k x^k \cdot \binom{k+2}{2}$, $\frac{1}{(1-4x)^4} = \dots$ etc.

useful for extracting coefficients of a rational function after performing the Partial fraction expansion.

⑥ $\frac{1}{\sqrt{1-4x}} = (1-4x)^{-1/2} = \sum_{k \geq 0} \binom{-1/2}{k} (-4)^k x^k$
 $= \sum_{k \geq 0} \left(\frac{(-1/2)(-3/2)(-5/2) \dots (-1/2 - (k-1))}{k!} (-4)^k \right) x^k$
 $= \sum_{k \geq 0} \left(\frac{2^k (1)(3)(5) \dots (2k-1)}{2^k k!} \right) x^k$
 $= \sum_{k \geq 0} \left(\frac{(2k)(1)(3)(5) \dots (2k-1)}{2^k k!} \right) x^k$
 $= \sum_{k \geq 0} \frac{(2k)!}{k! \cdot k!} x^k = \sum_{k \geq 0} \binom{2k}{k} x^k$
 central binomial coefficients... interesting...

this F.P.S.
 is algebraic
 but not rational

Another tool from calculus that's useful for $R[[X]]$

DEF'N For $A(x) = \sum_{n \geq 0} a_n x^n \in R[[X]]$, we define
 the formal derivative $A'(x) := \sum_{n \geq 1} \underbrace{n \cdot a_n}_{\substack{\text{"} \\ a_1 + 2a_2 + \dots + n a_n}} x^{n-1} \in R[[X]]$

The derivative satisfies the usual rules from calculus:

- $(A(x) + B(x))' = A'(x) + B'(x)$

- $(AB)' = A' \cdot B + B' \cdot A$

- $\left(\frac{1}{A}\right)' = \frac{-A'}{A^2}$

- $(A(B(x)))' = A'(B(x)) \cdot B'(x)$

- etc...